

Intelligent Codesign Strategy for Thermal Management and Cost Control of 3-D Integrated System With TTSV

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Abstract—The intelligent codesign strategy is proposed to balance the thermal management and cost control for 3-D integrated system with thermal through-silicon via (TTSV). The complex relationship between the thermal performance indexes and parameters of TTSV in the 3-D integrated system is investigated by the COMSOL Multiphysics software. According to the design logic, the row/column number of TTSV array and the distance of two TTSV unit are the global parameters, while the thickness of oxide layer, the radius of TTSV, and the height of oxide insulating layer are the local parameters. In order to improve the simulation efficiency, the modified neural network models are established based on the simulation data to replace the finite-element model. In the proposed codesign strategy, the thermal management and cost control are considered in the multiobjective evaluation function. According to the particle swarm optimization (PSO) algorithm, the multilevel PSO algorithm is developed to optimize the global and local parameters of TTSV. The effectiveness of the proposed codesign strategy is verified by six cases of two testbeds. The simulated temperature indexes are all lower than the desired targets (360, 320, and 20 K), and the number of TTSVs is decreased at the same time. Therefore, the proposed codesign strategy can not only realize the excellent thermal management but also achieve a compromise between thermal management and cost control.

Index Terms—3-D integrated system, cost control, multilevel optimization, thermal management, thermal through-silicon via (TTSV).

I. INTRODUCTION

CHIPLET can be considered as a little “IP” [1], [6]. Due to the advantages of low cost, high performance, modularization, and easy integration, the chiplet technology has been proposed to deal with the challenges caused by the failure of Moore’s law [7], [12]. Chiplet technology has been used in the high-performance computing system [13], [15] and memory system [16], [18]. With the increase of integration, the thermal management of chiplet-based system is an urgent problem. Chiplet-based system is a kind of complex 3-D integrated system, and its thermal management is more difficult. Thus, it is valuable to investigate the thermal management method of 3-D integrated system to further manufacture high-performance chiplet-based system.

In the previous research, some methods or technologies have been proposed to achieve the thermal management of 3-D integrated system, such as thermal through-silicon via (TTSV) [19], microchannel [20], and bond layer with high thermal conductivity [21]. In order to reduce the peak temperature (PT), Goplen and Sapatnekar [22] designed the TTSV for 3-D integrated system, and the simulated PT of 3-D integrated system with the designed TTSV can be reduced by 57% compared to the uniformly placed TTSV. Gagan et al. [23] designed and fabricated the new TTSV structure to absorb heat and harvest energy at the same time. Combined microchannel and TTSV, Shi et al. [24] proposed a hybrid cooling method for 3-D integrated system, which can achieve the excellent cooling capability. Based on the redistribution layer and TTSV array, Wang et al. [25] proposed a heat dissipation system for 3-D integrated system. The simulation results show that the PT is decreased by 3.8 °C. In these conventional design methods, the finite-element method (FEM)-based software or equivalent thermal conductivity models are utilized to determine the design parameters by iterative computation and comparison. However, these conventional design methods are low efficiency and rely on the experts’ experience. The high-efficiency design strategy

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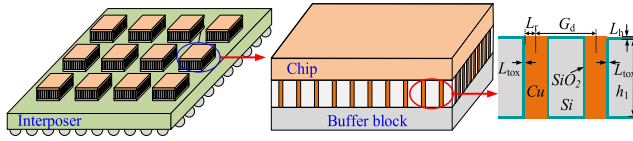


Fig. 1. Schematic of TTSV array in the 3-D integrated system.

for thermal management of 3-D integrated system should be investigated.

Recently, some optimization methods have been applied in the thermal management of 3-D integrated system, such as genetic algorithm (GA), simulated annealing algorithm, and particle swarm optimization (PSO) algorithm. Based on the simulated annealing algorithm, Ren et al. [26] proposed a hierarchical optimization method for 3-D integrated system to achieve thermal management, and the PT is reduced by 16 °C with a 3.5% increase in wire length. Based on the simulated annealing algorithm and random walk insertion algorithm, Wong and Lim [27] proposed a floor planning technology for 3-D integrated system with TTSV to achieve the 17% temperature reduction with 3% TTSV density. In addition, combining the PSO algorithm and equivalent thermal conductivity model, Wang et al. [28] proposed an optimization design method for the TSV array, which can achieve the thermal management of 3-D integrated system, and the design efficiency is improved by about 20%. These proposed methods are all focused on the thermal management of integrated system. Generally, the thermal dissipation performance is increased with increasing the number of TTSVs, but the cost is also increased. Therefore, the intelligent codesign strategy should be developed to balance the thermal management and cost control for 3-D integrated system.

Based on the GA-based back propagation neural network (GA-BPNN) and multilevel PSO algorithm, an intelligent codesign strategy is developed to balance the thermal management and cost control for 3-D integrated system with TTSV. The novelty of this research can be summarized as follows.

- 1) The GA-BPNN models are established based on simulation data to improve the design efficiency.
- 2) Under the framework of PSO algorithm, the multilevel optimization method is developed to optimize the global and local parameters of TTSV.
- 3) The proposed codesign method can effectively achieve the thermal management and cost control of 3-D integrated system.

In this article, the simulations and analyzation of thermal distribution in a 3-D integrated system with TTSV are shown in Section II. Section III gives the intelligent codesign strategy for 3-D integrated system with TTSV. Section IV presents the verification and discussion of the developed codesign strategy. The conclusion is summarized in Section V.

II. SIMULATION AND ANALYZATION OF THERMAL DISTRIBUTION IN 3-D INTEGRATED SYSTEM WITH TTSV

A. FEM Model of TTSV

The COMSOL multiphysics is adopted to simulate and analyze the thermal distribution in a 3-D integrated system with TTSV. The system model is chosen to investigate the

TABLE I
MATERIALS AND PARAMETERS UTILIZED IN THE FEM MODEL

Materials	Thermal conductivity (W/(m*K))	Density (kg/m ³)	Relative permittivity (F/m)	Heat capacity at constant pressure (J/(kg*K))
Cu	401	8960	1.0	384
SiO ₂	1.4	2200	4.2	730
Si	130	2329	11.7	700
Buffer block	500	5000	/	500

effect of TTSV parameters on thermal distribution, and the schematic of 3-D integrated system with TTSV array is shown in Fig. 1. The left of Fig. 1 is the schematic of hypothetical chiplet-based system, and its right is the specific model in this investigation. The function of the chips is memory or computing. According to the design logic, the row/column number of TTSV (G_{num}) and the distance of two TTSV units (G_d) are the global parameters, while the thickness of oxide layer (L_{tox}), radius of TTSV unit (L_r), and height of oxide insulating layer (L_h) are the local parameters. The design logic can be divided into four steps. First, the FEM model of TTSV array is established. Second, the design parameters of TTSV array are classified as global and local parameters. Third, the multilevel PSO algorithm is used to optimize the global and local parameters to achieve the balance between the thermal management and the cost control. Finally, the optimized global and local parameters are verified by finite-element simulation.

The materials and parameters utilized in the FEM model are presented in Table I. In the system, there are 12 chiplets, and however, two chiplets are selected to investigate the intelligent codesign strategy for thermal management and cost control. The other chiplets can be designed according to the developed method. In Testbed A, the power is 3×10^7 W/m², and the chip thickness is 10 μm . In Testbed B, the power is 4×10^7 W/m², and the chip thickness is 5 μm . In the simulations, both the width and length of chip are 300 μm , and only 1/4 system is simulated to decrease the computation. The height of TTSV unit is 50 μm . The power of heat source is imposed on the top surface of chip, and the heat flows from top to bottom. In the current research, because only two chiplets are investigated, the distance between each chiplet is not determined. The size of the block underneath the chip is equal to the size of chip. The bottom temperature of buffer block is 293.15 K, and the boundary conditions for edge are thermal insulation condition. The purpose of the buffer block is to smooth the heat flow that enters in and exists from the TTSV cell, and the value of the buffer block is flexible [29]. The bonding tier can be designed on the bottom of buffer block. The orthogonal design method is adopted to decrease the number of FEM simulation experiments.

B. Thermal Analyzation of TTSV

G_{num} and G_d are the global parameters, while L_r , L_{tox} , and L_h are the local parameters. The ranges of G_{num} and G_d are [4, 7] and [12, 18] μm , while the ranges of L_r , L_{tox} , and L_h are

TABLE II
ORTHOGONAL DESIGN OF GLOBAL PARAMETERS

Case number	G_{num}	G_d	Case number	G_{num}	G_d
1	6	18	9	6	14
2	5	16	10	7	16
3	6	12	11	7	18
4	4	18	12	7	14
5	5	14	13	4	12
6	4	14	14	6	16
7	5	12	15	4	16
8	7	12	16	5	18

TABLE III
ORTHOGONAL DESIGN OF LOCAL PARAMETERS

Local case	L_r (μm)	L_{tox} (μm)	L_h (μm)	Local case	L_r (μm)	L_{tox} (μm)	L_h (μm)
1	3	0.1	3	14	4	0.3	5
2	5	0.1	5	15	3	0.3	4
3	4	0.1	4	16	4	0.2	2
4	4	0.5	1	17	1	0.5	3
5	5	0.4	4	18	2	0.4	1
6	1	0.1	1	19	3	0.4	2
7	5	0.5	2	20	4	0.4	3
8	1	0.4	5	21	1	0.2	4
9	2	0.2	5	22	2	0.5	4
10	2	0.1	2	23	3	0.5	5
11	1	0.3	2	24	5	0.2	3
12	5	0.3	1	25	2	0.3	3
13	3	0.2	1	/	/	/	/

[1, 5], [0.1, 0.5], and [1, 5] μm , respectively. The performance indexes include PT, average temperature (AT), and temperature difference (DT) of the bottom surface of chip. The orthogonal design results of global and local parameters are shown in Tables II and III. Based on the orthogonal design, the number of global cases is 16, and the number of local cases is 25. Thus, the number of total cases for finite-element simulation is 400. The simulation results of TTSV array with different design parameters are shown in Fig. 2. In Testbed A, the ranges of PT, AT, and DT are [314.11, 364.47], [301.06, 328.32], and [3.47, 27.66] K, respectively. In Testbed B, the ranges of PT, AT, and DT are [324.5, 546.45], [304.13, 414], and [6.09, 102.22] K, respectively. According to the simulation results, the global and local parameters affect the thermal distribution of 3-D integrated system. Obviously, when the num is 7, the PT and AT are relatively low. The thermal distribution is varied with the global and local parameters, and the thermal distribution is complex and irregular. Under the requirement of cost, the thermal management of 3-D integrated system is a complex optimization problem.

III. INTELLIGENT CODESIGN STRATEGY FOR 3-D INTEGRATED SYSTEM WITH TTSV

The intelligent codesign strategy is proposed to balance the thermal management and cost control of 3-D integrated system with TTSV. In the proposed strategy, the GA-BPNN models are established based on the simulation data to improve the simulation efficiency. The multiobjective evaluation function is established considering the thermal management and cost control. Under the framework of PSO algorithm, the multilevel optimization algorithm is developed to optimize the global and local parameters of TTSV array.

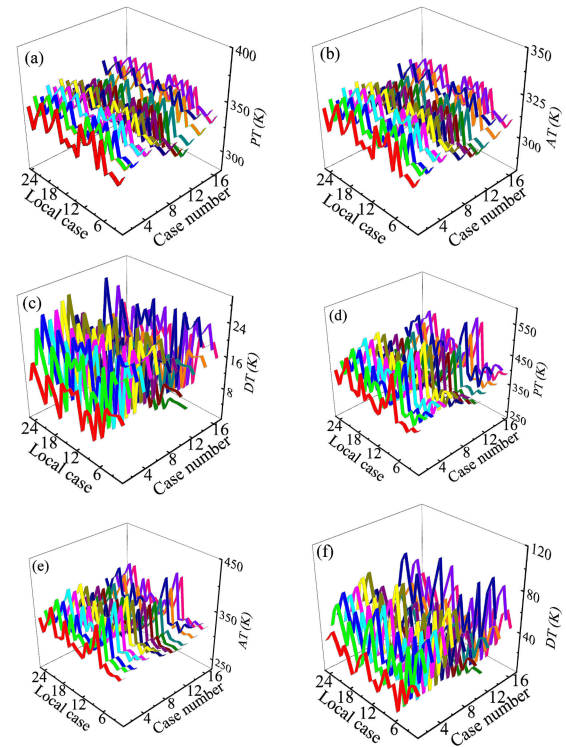


Fig. 2. Simulation results of finite-element model at global and local parameters: (a) PT of Testbed A, (b) AT of Testbed A, (c) DT of Testbed A, (d) PT of Testbed B, (e) AT of Testbed B, and (f) DT of Testbed B.

A. GA-BPNN Model

NN has the advantages of self-learning, self-adaptation, and self-organization, and it has been developed and utilized in the modeling and optimization of complex engineering problems. The NN model is used to describe the mapping relationship between design parameters and thermal distribution indexes. The simulation data of design parameters and thermal distribution indexes are utilized for training the NN models. In the established NN model, the inputs are the global and local parameters, and the outputs are temperature indexes. Generally, the more the amount of data, the higher the accuracy of the trained NN model. This is because the sufficient data can describe the characteristic of the system in detail. In addition, the accuracy of NN is affected by the number of hidden layers and the number of nodes in the hidden layer. If the number of hidden layers is too small and the feature is trained insufficiently, the accuracy will be low. The number of nodes in the hidden layer also affects the accuracy of NN. The accuracy of NN model is first increased and then decreased with the increase of the node number in the hidden layer [30]. According to Kolmogorov's theorem [31], the node number of hidden layer in the trained GA-BPNN model is 11. The number of datapoints for training GA-BPNN model is 400, and the total training time is less than 28 s. The NN model is expressed as follows:

$$\text{PT/AT/DT} = g(W_2 \cdot g(W_1 U + b_1) + b_2) \quad (1)$$

$$g(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

where $g(x)$ is the activation function. The inputs and outputs of NN are U and PT/AT/DT, respectively. W_1 and W_2 are

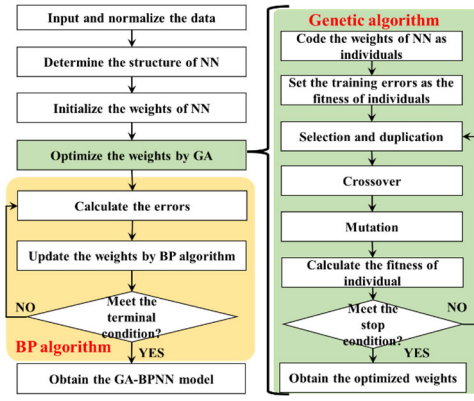


Fig. 3. Flowchart of GA-BPNN model.

the weight matrices, respectively. b_1 and b_2 represent the bias terms, respectively.

The NN model is generally trained by the BP algorithm. The accuracy of BPNN model is affected by the initial weights because the weights determined by the BP algorithm are easily convergence to the local optimal solution [32]. In order to overcome this problem, GA is used to determine the initial weights of NN, and its flowchart is shown in Fig. 3.

B. Multiobjective Evaluation Function for Thermal Management and Cost Control

The multiobjective evaluation function is constructed considering the thermal management and cost control of 3-D integrated system. The thermal management can be achieved by controlling the temperature indexes (PT, AT, and DT), and the cost control can be achieved by controlling the row/column number of TTSV. Thus, the multiobjective evaluation function is the weighted sum of PT, AT, DT, and G_{num} , which can be expressed as follows:

$$J = \alpha \cdot (PT - PT_{\text{des}})^2 + \beta \cdot (AT - AT_{\text{des}})^2 + \lambda \cdot (DT - DT_{\text{des}})^2 + \gamma \cdot G_{\text{num}}^2 \quad (3)$$

where PT_{des} , AT_{des} , and DT_{des} are the desired PT, AT, and DT, respectively, and α , β , λ , and γ represent the weight coefficients of PT, AT, DT, and G_{num} , respectively. The magnitude effect of performance indexed can be avoided by normalization, the normalized multiobjective criteria can be expressed as follows:

$$J = \alpha \cdot \left(\frac{PT - PT_{\text{des}}}{PT_{\text{max}} - PT_{\text{min}}} \right)^2 + \beta \cdot \left(\frac{AT - AT_{\text{des}}}{AT_{\text{max}} - AT_{\text{min}}} \right)^2 + \lambda \cdot \left(\frac{DT - DT_{\text{des}}}{DT_{\text{max}} - DT_{\text{min}}} \right)^2 + \gamma \cdot \left(\frac{G_{\text{num}}}{\text{num}_{\text{max}} - \text{num}_{\text{min}}} \right)^2 \quad (4)$$

where PT_{max} , AT_{max} , DT_{max} , and num_{max} are the maximum of PT, AT, DT, and G_{num} , respectively. PT_{min} , AT_{min} , DT_{min} , and num_{min} are the minimum of PT, AT, DT, and G_{num} , respectively.

C. Multilevel PSO Algorithm

In this research, the design parameters include the global and local parameters, which means that the solutions belong to different levels, and the global and local solutions cannot

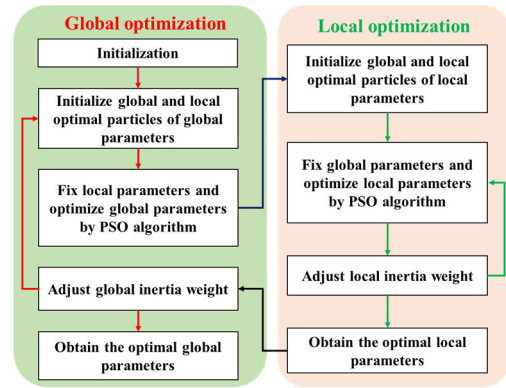


Fig. 4. Flowchart of multilevel PSO algorithm with LDIW for TTSV.

TABLE IV
PARAMETERS OF THE MULTIOBJECTIVE EVALUATION FUNCTION

Testbed	Case	Desired performance			Weight coefficients			
		PT (K)	AT (K)	DT (K)	α	β	λ	γ
A	A	320	305	10	0.4	0.3	0.3	0
	B	320	305	10	0.3	0.2	0.2	0.3
	C	340	315	15	0.3	0.2	0.2	0.3
B	A	340	315	15	0.4	0.3	0.3	0
	B	340	315	15	0.3	0.2	0.2	0.3
	C	360	325	20	0.3	0.2	0.2	0.3

be separately optimized. However, the standard optimization algorithm cannot effectively optimize the global and local parameters for TTSV in the 3-D integrated system. The PSO algorithm is an effective evolutionary computation method. Due to the advantages of fast convergence and easy to implement, the PSO algorithm has been used in the optimization design of electronic components. Thus, the PSO algorithm can be used in this research to optimize the design parameters of TTSV to achieve the balance between the thermal management and the cost control. Under the framework of PSO algorithm with linearly decreasing inertia weight (LDIW), the multilevel optimization algorithm is developed to optimize the global and local parameters of TTSV in the 3-D integrated system. The flowchart of multilevel PSO algorithm with LDIW for TTSV is shown in Fig. 4, and it can be summarized as follows.

Step 1: Initialize the parameters of optimization algorithm and optimality criterion.

Step 2: Optimize the global parameters. The local parameters are fixed, and the global parameters are optimized according to the multiobjective evaluation function.

Step 3: Optimize the local parameters. Fix the optimized global parameters, and the local parameters are optimized according to the multiobjective evaluation function.

Step 4: The inertia weight is adjusted until the optimal local parameters are obtained in the current iteration.

Step 5: The inertia weight is adjusted until the optimal global parameters are obtained.

IV. VERIFICATION AND DISCUSSION

Two testbeds are conducted to verify the effectiveness of the developed intelligent codesign strategy for thermal management and cost control of 3-D integrated system with TTSV.

TABLE V
PARAMETERS OF THE PROPOSED INTELLIGENT CODESIGN STRATEGY

Parameters of multi-level PSO algorithm	Global parameters	Constant parameters	$c_1^g = 2, c_2^g = 2$
		Range of inertia weight	$w^g \in [0.4, 0.9]$
		Maximum generation	$MaxG^g = 20$
		Population size	$P^g = 50$
		Range of particle position	$x_{num}^g \in [4, 7], x_d^g \in [12, 18]$
		Range of particle velocity	$v_{num}^g \in [-10, 10], v_d^g \in [-20, 20]$
	Local parameters	Constant parameters	$c_1^l = 2, c_2^l = 2$
		Range of inertia weight	$w^l \in [0.4, 0.9]$
		Maximum generation	$MaxG^l = 30$
		Population size	$P^l = 80$
		Range of particle position	$x_t^l \in [1, 5], x_{tox}^l \in [0.1, 0.5], x_h^l \in [1, 5]$
		Range of particle velocity	$v_t^l \in [-5, 5], v_{tox}^l \in [-1, 1], v_h^l \in [-5, 5]$

TABLE VI
OPTIMIZED PARAMETERS FOR THREE CASES

Testbed	Case	Global parameters		Local parameters		
		G_{num}	$G_d (\mu m)$	$L_r (\mu m)$	$L_{tox} (\mu m)$	$L_h (\mu m)$
A	A	7	17.26	4.03	0.3	1
	B	5	18	5	0.1	5
	C	4	18	3.85	0.5	1
B	A	7	17.3	2.13	0.1	1
	B	4	18	1.51	0.1	1
	C	4	18	2.49	0.21	1

A. Codesign Plan for 3-D Integrated System With TTSV

The parameters of the multiobjective evaluation function for three cases of two testbeds are shown in Table IV. The parameters of the multilevel PSO algorithm are shown in Table V, which include global and local parameters. In Case A, only the thermal management is considered, and the desired PT, AT, and DT are relatively low, which implies the high thermal management requirement. The thermal management and cost control are considered in Cases B and C. The desired PT, AT, and DT of Case B are equal to Case A, and this case is to verify the thermal management effect under the cost control requirement. In Case C, the desired PT, AT, and DT are higher than those of Case B, and this case is to verify the compromise performance between thermal management and cost control. According to the codesign plan for 3-D integrated system with TTSV, the proposed intelligent codesign strategy is conducted to verify its effectiveness.

B. Verification

In two testbeds, the optimized global and local parameters of TTSV are presented in Table VI. Because only the thermal management is considered in Case A, the global parameter G_{num} of Case A is the largest, which is beneficial for heat dissipation. The cost is considered in Cases B and C, so the global parameter G_{num} of Case B/C is small, which is beneficial to decrease the cost.

In addition, the simulation results of 1/4 system at the optimized parameters based on two testbeds are shown in

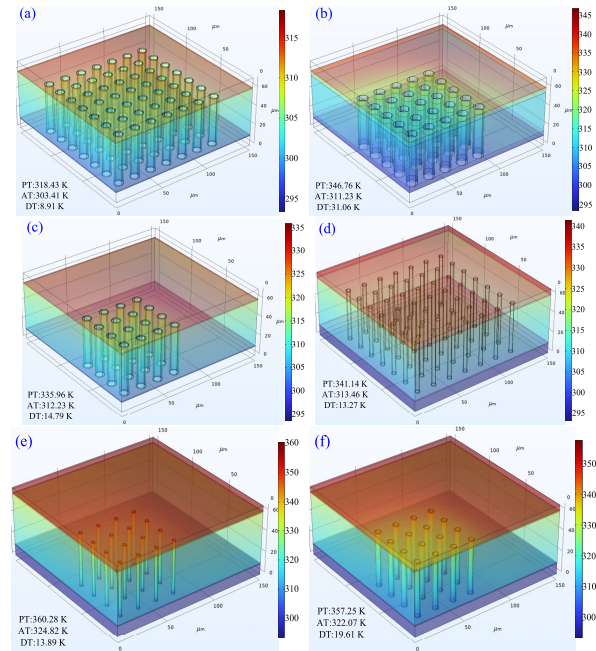


Fig. 5. Thermal distributions: (a)–(c) Cases A–C of Testbed A and (d)–(f) Cases A–C of Testbed B.

Fig. 5. Obviously, the temperature is low in the region distributed TTSV, while the temperature of the marginal region is high. The temperature curves of diagonal line are shown in Fig. 6. The diagonal line is located on the bottom surface of chip, which is $Z = 60 \mu m$. The diagonal line is a line segment from point (0, 0, 60) to point (150, 150, 60). The trend of temperature is increased from the center to the marginal region. In Testbed A, the PTs of Cases A–C are 318.43, 346.76, and 335.96 K, respectively. Their minimum temperatures are 309.52, 315.70, and 321.17 K. In Testbed B, the PTs of Cases A–C are 341.14, 360.28, and 357.25 K, respectively. Their minimum temperatures are 327.87, 346.39, and 337.64 K. Clearly, the DTs of Cases A and C in Testbed A are relatively small, but the DT of Case B is relatively large, which implies that the temperature distribution of Case B is not uniform.

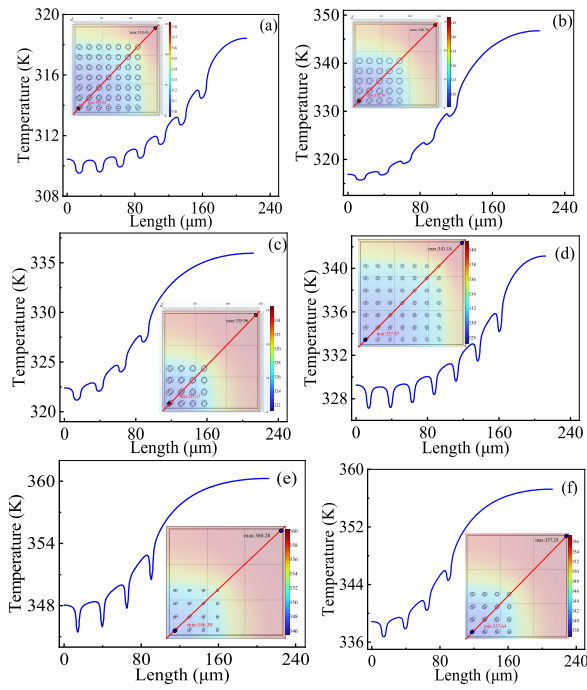


Fig. 6. Temperature curves of diagonal line: (a)–(c) Cases A–C of Testbed A and (d)–(f) Cases A–C of Testbed B.

TABLE VII
DESIRED, OPTIMIZED, AND SIMULATED TEMPERATURE
INDEXES OF DIFFERENT CASES

Testbed	A			B		
Indexes	PT (K)	AT (K)	DT (K)	PT (K)	AT (K)	DT (K)
Desired	A 320	305	10	340	315	15
	B 320	305	10	340	315	15
	C 340	315	15	360	325	20
Optimized	A 321.14	304.22	9.97	341.51	313.02	13.36
	B 330.86	307.15	19.89	360.24	315.8	14.53
	C 346.40	316.36	18.93	357.57	323.96	19.06
Simulated	A 318.43	303.41	8.91	341.14	313.46	13.27
	B 346.76	311.23	31.06	360.28	324.82	13.89
	C 335.96	312.23	14.79	357.25	322.07	19.61

The desired, optimized, and simulated temperature indexes for Testbeds A and B are shown in Table VII. The values of desired temperature indexes are determined based on extensive simulation analysis and project requirements.

In Testbed A, the relative error of Case A between the simulated and desired PTs/ATs is 0.49%/0.52%, and the difference between the desired and simulated DTs is 1.09 K. In Testbed B, the relative error of Case A between the simulated and desired PTs/ATs is 0.34%/0.49%, and the difference between the desired and simulated DTs is 1.73 K. The temperature indexes obtained at the optimized design parameters well agree with the desired targets, so the developed method can effectively achieve the thermal management of 3-D integrated system with a TTSV array. In Case B of Testbed A, the optimized/simulated PT, AT, and DT are all larger than the desired targets, so the thermal management is not achieved under the consideration of cost control. In Case B of Testbed B, the optimized/simulated PT and AT are larger than the desired ones, so the thermal management is also not achieved under the consideration

of cost control. This is because the temperature indexes of Case B are too harsh for two testbeds. The optimized G_{num} of Testbed A is 5, while Testbed B is 4, so the heat dissipation effect of Case B in two testbeds is worse than that of Case A ($G_{\text{num}} = 7$). In Case C, the requirement of thermal management of two testbeds is decreased. The desired PT, AT, and DT of Testbed A are 340, 315, and 15 K, respectively. The simulated PT, AT, and DT are 335.96, 312.23, and 14.79 K, respectively, which are all smaller than the desired targets. In addition, the optimized G_{num} of Case C is 4, which implies that the cost is very low. In Testbed B, the simulated PT, AT, and DT of Case C are 357.25, 322.07, and 19.61 K, respectively, which are smaller than the desired ones (360, 325, and 20 K). Therefore, the developed codesign method can achieve a compromise between thermal management and cost control of 3-D integrated system with TTSV.

In Testbed A, the difference between the optimized and simulated results for Case B/C is relatively large compared to Case A. In Case A, only the thermal management is considered in the multiobjective evaluation function, so the global parameter G_{num} of Case A is the largest, which is beneficial for heat dissipation. In Cases B and C, the thermal management and cost control are considered in the multiobjective evaluation function, so the global parameter G_{num} is smaller than Case A, which implies that the heat dissipation ability is lower than Case A ($G_{\text{num}} = 7$). According to the optimized and simulated results of Testbed A, it can be seen that the optimized temperatures in Cases B and C are underestimated and overestimated compared with the simulated ones. This is because the desired targets are different between Cases B and C, and the temperature indexes of Case B are too harsh. In addition, the relative error is existed in the trained GA-BPNN model, and the optimized results are varied for different desired targets. In Testbed A, the optimized L_h is 5 in Case B, which is more expensive than a lower L_h value, but the thermal management and cost control are considered in Case B. Although decreasing the row/column number of TTSV can decrease the cost, the thermal management cannot be achieved.

In addition, the nondominated sorting GA-II (NSGA-II) and PSO algorithm with linear decreasing inertia weight (PSO-LDIW) is adopted to optimize the design parameters of TTSV array for Testbed B. The optimized and simulated results of the PSO-LDIW and NSGA-II algorithms are shown in Table VIII. The iteration numbers of three optimization algorithms are equal. The desired temperature indexes of three cases are equal to the desired target of Testbed B. In Case C, the optimized and simulated PTs of PSO-LDIW algorithm are 368.58 and 370.45 K, and the optimized and simulated ATs of PSO-LDIW algorithm are 329.74 and 330.64 K, which are all larger than the desired targets (360 and 325 K). In addition, the optimized and simulated PTs of NSGA-II algorithm are 362.48 and 363.08 K, and the optimized and simulated DTs of NSGA-II algorithm are 20.81 and 20.9 K, which are all larger than the desired targets (360 and 20 K). However, the optimized and simulated results of Case C using the multilevel PSO algorithm are all smaller than the desired targets. Therefore, the proposed multilevel PSO algorithm can effectively balance the different requirements to obtain the optimal parameters.

TABLE VIII
OPTIMIZED AND SIMULATED RESULTS OF PSO-LDIW AND NSGA-II ALGORITHMS

Algorithm		G_{num} (μm)	G_d (μm)	L_r (μm)	L_{tox} (μm)	L_h (μm)	Optimized			Simulated		
							PT (K)	AT (K)	DT (K)	PT (K)	AT (K)	DT (K)
PSO-LDIW	A	7	18	5	0.5	5	338.02	307.16	22.15	337.08	307.06	23.15
	B	4	12	1	0.1	1	362.98	327.38	10.13	363.33	327.51	10.45
	C	4	12	1.15	0.12	1.15	368.58	329.74	13.05	370.45	330.64	13.45
NSGA-II	A	7	18	2.79	0.39	1.28	341.97	313.76	14.72	342.49	313.67	13.85
	B	4	16.89	2.83	0.25	1	356.76	321.01	21.86	356.57	321.31	22.23
	C	4	17.98	2.52	0.34	1.13	362.48	324.49	20.81	363.08	324.71	20.9

TABLE IX
COMPARISON RESULTS WITH THE TRADITIONAL METHOD

Method	PT (K)	AT (K)	DT (K)	Time (s)
Developed	357.25	322.07	19.61	2
[25]	390.73	/	/	/
[27]	355.75	/	/	463

C. Discussion

The thermal management of 3-D integrated system is a complex optimization problem. In addition, the thermal management coupled with cost control makes the design of 3-D integrated system more complex. In the traditional design method, the design parameters are determined through iterative computation and comparison using FEM software, so a large amount of FEM simulations should be conducted to find the relatively optimal results based on the experience of designer, and this result may not be the optimal results. Generally, the temperature indexes are different for different requirements, and the run times of different simulation models are different. In [25], finite-element software is used to simulate the eight-layer integrated circuit system model, and the obtained PT is 390.73 K. In the developed method, the PT, AT, and DT of the bottom surface of chip for Testbed B are 357.25, 322.07, and 19.61 K, respectively. In [27], the obtained PT by using the traditional method is 355.75 K, and however, the run time of single simulation is 463 s, which is much larger than that of the developed method (2 s). The comparison results are shown in Table IX. Thus, the design effectiveness and efficiency of the developed method are clearly improved.

In the proposed intelligent codesign strategy, the GA-BPNN model is used to replace the FEM software to improve the run time, and the PSO algorithm can effectively determine the optimal design parameters without designers according to the desired targets. The run time of the established GA-BPNN model is less than 2 s, which is much smaller than that of FEM software (>140 s). Indeed, during the establishment of GA-BPNN models, the FEM simulation data obtaining (400 iterations of FEM software) and GA-BPNN training take time. However, the traditional design method requires uncertain iterations of FEM simulation, and the iteration number or design cycle is dependent on the experience of designers. In the proposed method, the iteration number of FEM simulation is certain, and the optimal results can be automatically obtained with low time cost. The run time of the proposed intelligent codesign method for one independent run is less than 1200 s, which can effectively reduce the design cycle compared with the traditional method.

The proposed codesign strategy can effectively improve the heat dissipation capability of TTSV and decrease the cost of 3-D integrated system. However, the errors between the optimized and simulated results have existed. The optimized results are determined by the GA-BPNN model. Due to the finite data, the trained NN model cannot very precisely describe the complex and irregular relationship between the parameters and temperature indexes. The optimized temperatures are higher than the desired ones. This is because the relative error exists in the trained GA-BPNN model. In addition, the balance between the thermal management and the cost control is hard to be achieved during the optimization. Under the requirement of cost control, the optimized temperatures may be higher than the desired ones. In Testbed A, reducing the number of TTSV leads to higher temperature from Cases A to B but leads to a decrease from Cases B to C. This is because the desired targets are different among the three cases. In Case A, only the thermal management is considered in the multiobjective evaluation function, so the global parameter G_{num} of Case A is the largest, which is beneficial for heat dissipation. In Case B, the thermal management and cost control are considered in the multiobjective evaluation function, so the global parameter G_{num} is smaller than Case A, which implies that the heat dissipation ability is lower than Case A. In addition, the desired targets are different between Cases B and C. The temperature indexes of Case B are too harsh to be achieved, while the temperature indexes can be achieved in Case C.

In our future research, the large database will be established for 3-D integrated system to improve the design accuracy and reliability. In addition, the cost is simply represented by the number of TTSVs, which is insufficiently rigorous. The rationality of the multiobjective evaluation function will be investigated, and the evaluation mechanisms will be established. The KOZs will be investigated for multiphysics coupling design of TTSV.

V. CONCLUSION

In this research, based on the GA-BPNN model and PSO algorithm, an intelligent codesign strategy is proposed to balance the thermal management and cost control for 3-D integrated system with TTSV. The proposed codesign strategy can effectively achieve the excellent heat dissipation capability of TTSV and decrease the cost of 3-D integrated system. The conclusions are given as follows.

- 1) The complex relationship among global, local parameters, and thermal performance indexes of 3-D integrated system with TTSV is investigated, and the GA-BPNN

model is established based on the simulation data to rapidly describe this relationship.

- 2) The multiobjective evaluation function considering the thermal management and cost control for 3-D integrated system is established, and the multilevel PSO algorithm is proposed to optimize the global and local parameters of TTSV.
- 3) The proposed codesign strategy can not only realize the excellent thermal management but also achieve a compromise performance between thermal management and cost control. In addition, the proposed codesign strategy can effectively improve the design efficiency.

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